# Type MS · Identify the independent or dependent variable

## Answer 1 of 10 · Question 4 part b.ii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 4 (a) (i) | Any three of the following:<br>• add alcohol/ethanol to food;<br>• shake;<br>• add water;<br>• milky coloration shows fat present; | Allow rub food;<br>onto filter paper;<br>translucent;<br>answer must be in sequence i.e. ethanol first | 3 |
| (ii) | lipase | | 1 |
| (iii) | fatty acids;<br>glycerol; | | 2 |
| (b) (i) | Any value between 8.6 and 8.8; | | 1 |
| (ii) | independent = time/ temperature;<br>dependent = pH; | | 2 |
| (iii) | • 37°C is the optimum temperature;<br>• lipase/enzyme is more active/works faster;<br>• fat is digested more quickly;<br>• at 80 °C/higher temperature enzyme becomes denatured; | Allow reverse argument throughout:<br>• at 80°C/higher temperature enzyme is denatured;<br>• enzyme less active;<br>• fat digested more slowly;<br>• 37°C is the optimum temperature (for enzyme activity/enzyme works faster at 37°C | 4 |

**Total marks 13**

## Type MS · Identify the independent or dependent variable

### Answer 1 of 10 · Question 4 part b.ii · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Identify the independent or dependent variable

### Answer 2 of 10 · Question 8 part b.i · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 8 (a)(i) | • high (blood) pressure (in the glomerulus);<br>• small molecules/named small molecule forced / filtered/pass through (into Bowmans capsule); | Do not allow protein | 1<br>1 |
| (ii) | Any four from:<br>• lower concentration of glucose/oxygen in blood in renal vein;<br>• used in respiration;<br>• less/no urea in renal vein;<br>• urea excreted in urine;<br>• more carbon dioxide in renal vein;<br>• carbon dioxide produced in respiration; | Allow reverse argument throughout | Max 4 |
| (b) (i) | quantity of salt (in food); | | 1 |
| (ii) | Any two from:<br>• different foods used (which may affect the amount of urea produced/amount of water reabsorbed);<br>• different people used who may be different genders/ages/ have a medical condition/different metabolism (that affects osmoregulation/water balance);<br>• the food/drink consumed by the people prior to the investigation was not taken into account (which will influence the quantity of urea/water in urine produced); | | Max 2 |
| (iii) | use one person only (and vary salt intake)/monitor / control food/water consumption/give each person the same food (but vary salt content); | Allow valid alternatives | 1 |

**Total question 8 = 10 marks**

## Type MS · Identify the independent or dependent variable

### Answer 2 of 10 · Question 8 part b.i · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Identify the independent or dependent variable

### Answer 3 of 10 · Question 4 part a.iii · 1 mark

2020 NOV · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 4 (a) | (i) | • axes labelled (1)<br>• suitable scale (1)<br>• axes correct way round (1)<br>• correct plots (1)<br>• lines labelled/key given (1) | | 5 |
| | (ii) | any two from<br>• volume of water (1)<br>• mass of carbohydrate/aspirin (1)<br>• temperature (1)<br>• time interval between each reading (1) | Allow type of aspirin<br>Allow amount as an alternative to mass<br>Ignore concentration | 2 |
| | (iii) | pH (1) | | 1 |
| (b) | | • (aspirin) dissolves/reduces pH more quickly (1)<br>• (when combined) with sucrose (1)<br>• so will act more quickly/relieves pain faster (1) | | 3 |

Total 11 marks

## Type MS · Identify the independent or dependent variable

### Answer 3 of 10 · Question 4 part a.iii · 1 mark

2020 NOV · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

## Type MS · Identify the independent or dependent variable

### Answer 4 of 10 · Question 7 part a.ii · 1 mark

2021 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 7 (a) | (i) | • temperature of incubation;<br>• volume of milk;<br>• composition of nutrients in agar/type of agar; | | 3 |
| | (ii) | size of bacterial growth; | | 1 |
| (b) | (i) | $\frac{7.1}{5} = 1.42$;<br><br>$\frac{36.5}{5} = 7.3$; | One mark for each correct answer | 2 |
| | (ii) | • more bacteria in B;<br><br>• at start of investigation;<br>• because extra added to tube;<br>• some bacteria in A/milk which grew; | Allow more growth in B/growth increases more quickly in B<br><br>Allow reference to contamination e.g.from air | 4 |
| (c) | | • transfer loop sterilised/heated in flame;<br><br>• transfer performed quickly; | Allow reference to sterilised in context / sterile technique / sterilise equipment / named equipment | 2 |
| (d) | | • repeat investigation;<br>• using other organisms; | | 2 |

Total 14 marks

## Type MS · Identify the independent or dependent variable

### Answer 4 of 10 · Question 7 part a.ii · 1 mark

2021 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Identify the independent or dependent variable

Answer 5 of 10 · Question 6 part a · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

| | | | |
|---|---|---|---|
| 6 (a) | (type of) antibacterial cleaner; | | 1 |
| (b) | Any two from the following<br>• use sterilised equipment/named equipment;<br>• heat the inoculating loop (in a Bunsen flame);<br>• do not open the lid of the Petri dish too much;<br>• seal the lid of the Petri dish; (with tape) | Named equipment must be hardware | 2 |
| (c) | • to prevent growth/reproduction;<br>• of pathogen/harmful microorganism; | | 2 |
| (d) | it is a control; | | 1 |
| | | | Total 6 |

## Type MS · Identify the independent or dependent variable

Answer 5 of 10 · Question 6 part a · 1 mark

2022 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Identify the independent or dependent variable

Answer 6 of 10 · Question 5 part a.ii · 1 mark

2024 JUN · Paper 01 · find part a.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 5 (a) | (i) | any five from<br><br>• place substrate into a test/boiling tube/beakers;<br>• place test tube in a (hot) water bath;<br><br>• add enzyme to substrate;<br>• measure time taken for reaction to occur;<br><br>• use same volume/concentration of enzyme/substrate;<br>• repeat for other temperatures; | Allow a temperature value<br><br>Allow example of a reaction e.g. colour change, amount of glucose produced<br><br>reject amount | (5) |
| | (ii) | temperature; | | (1) |
| (b) | | • less kinetic energy;<br>• less/fewer collisions (with substrate);<br>• less enzyme-substrate complexes formed/less binding;<br>• lower enzyme activity/rate of reaction; | | (4) |

**Total for question 5 = 10 marks**

## Type MS · Identify the independent or dependent variable

### Answer 7 of 10 · Question 2 part c · 1 mark

2024 NOV · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

| Question number | | | | | |
|---|---|---|---|---|---|
| 2 | (a) | (i) | • axes labels;<br>• plots;<br>• scale; | axes correct way round<br>use aty least half of the graph paper for the plotted results<br>ignore extrapolation of line | 3 |
| | | (ii) | line of best fit; | | 1 |
| | | (iii) | As time increases pH decreases; | ECF from an incorrectly drawn graph | 1 |
| | (b) | | Any three from:<br>• lipase;<br>• digests/breaks down fat (in milk);<br>• to fatty acids;<br>• which lowers pH/increases acidity; | | 3 |
| | (c) | | time; | allow mins/minutes | 1 |
| | (d) | | Any two from:<br>• temperature;<br>• volume/concentration of enzyme/substrate (solution)/volume of milk;<br>• type of milk; | ignore amount | 2 |
| | (e) | | repeat the experiment | | 1 |

**Total Marks 12**

## Type MS · Identify the independent or dependent variable

### Answer 7 of 10 · Question 2 part c · 1 mark

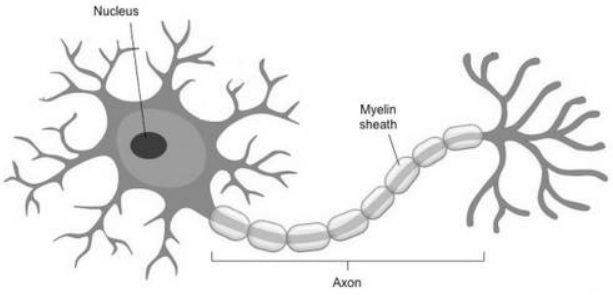
2024 NOV · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Identify the independent or dependent variable

### Answer 8 of 10 · Question 3 part b.ii · 1 mark

2025 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 3 | (a) | (i) |  | 1 mark for diagram 3 marks for labels | 4 |
| | | (ii) | • transmits electrical/nerve impulses;<br>• from CNS/brain/spinal cord/relay neurone, to effector organs/named effector organ; | | 2 |
| | (b) | (i) | Any six from<br>1 (student X) holds up the ruler between the thumb and forefinger (of other student);<br>2 then drops/releases the ruler;<br>3 (other) student catches the ruler;<br>4 record the reading on the ruler/distance fallen when caught;<br>5 repeat for <u>this</u> student;<br>6 using the same distance between thumb and forefinger;<br>7 find mean/average value of reading on ruler;<br>8 calculate reaction time (using formula);<br>9 repeat with <u>other</u> students; | | 6 |
| | | (ii) | the reading on the ruler/ reaction time; | | 1 |
| | | (iii) | independent variable not continuous/is discrete; | | 1 |
| | | (iv) | reaction time = √(0.15 ÷ 5);<br>reaction time = 0.17 (s); | accept 0.173(2) | 2 |
| | | | | **Total for Question 3 = 16 marks** | |

## Type MS · Identify the independent or dependent variable

### Answer 9 of 10 · Question 7 part b · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 7 (a) | • body temperature;<br>• optimum/enzyme is not denatured; | Accept alimentary for MP1 | 2 |
| (b) | • effect of pH/to see if pH affects enzyme activity;<br>• range of pH used/different pH used/ pH is the independent variable; | ***Effect of different pH on enzyme activity = 2 marks*** | 2 |
| (c) | Any four from:<br><br>Answers about validity( MP1-4)<br>• The same temperature;<br>• The same volume of enzyme/suspension;<br>• not (valid) because different time used;<br>• the same enzyme is used;<br><br>Answers about accuracy( MP 5-7)<br>• as difficult to judge endpoints/ when it goes clear;<br>• students' judgement of endpoint may vary/ subjective/ not standardised/ did not use a colorimeter;<br>• timing is too short to reach endpoint/ cannot tell it will go clear if left for longer; | They do not have to mention validity/accuracy to score marks.<br><br>Must have at least one mark from each section to score 4. Maximum of 3 if answers are from one section | 4 |
| (d) | • tube with no enzyme/ denatured enzyme/boiled enzyme;<br>• same volume of boiled enzyme/water /5cm$^3$ of water/boiled enzyme;<br>• same volume of suspension/same overall volume/ same volume of substrate ( in test tube); | | 3 |
| (e) | • optimum pH;<br>• fastest between 2 and 4; | | 2 |
| (f) | • stomach;<br>• acidic/low pH/pH2-4/pH2; | | 2 |

**Total 15 marks**

## Type MS · Identify the independent or dependent variable

### Answer 9 of 10 · Question 7 part b · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

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## Type MS · Identify the independent or dependent variable

Answer 10 of 10 · Question 5 part a.i · 1 mark

2024 JUN · Paper 02 · find part a.i in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 5 | (a) (i) | **C** the type of antibiotic added<br><br>The following are incorrect because:<br><br>A the mass of bacteria added is irrelevant<br>B the size of the inhibition zone is the dependent variable<br>D the mass of agar in the Petri dish is irrelevant | | (1) |
| | (ii) | Any two from:<br><br>• sterilise equipment/named equipment/disinfect/swab desk;<br>• use of sterile agar;<br>• reference to use of lid on Petri dish/ minimising the time the lid is removed;<br>• Flame from Bunsen burner: | | (2) |
| | (iii) | (antibiotic) Q;<br>greater inhibition zone /greater clear zone/ more bacteria killed; | | (2) |
| | (iv) | Any four from:<br><br>• mutation;<br><br>• (resistant) bacteria survive;<br><br>• reproduce;<br><br>• pass on resistant gene/allele;<br><br>• Increase of resistant gene/allele in the population/ increase in allele frequency (over many generations) | | (4) |
| | (b) | • (bacteria/fungi/decomposers) release enzymes/named enzyme<br><br>• breakdown dead/organic matter/ named organic matter;<br>• release nutrients/named nutrients into the soil/environment; | Ignore carbon dioxide<br>Ignore fertiliser | (2) |

**Total for Question 5 = 11 marks**